Agenda

Forms and Validations

- Angular Forms
- Forms Control and States
- State Transition
- Template Driven Form
- Model Driven Form

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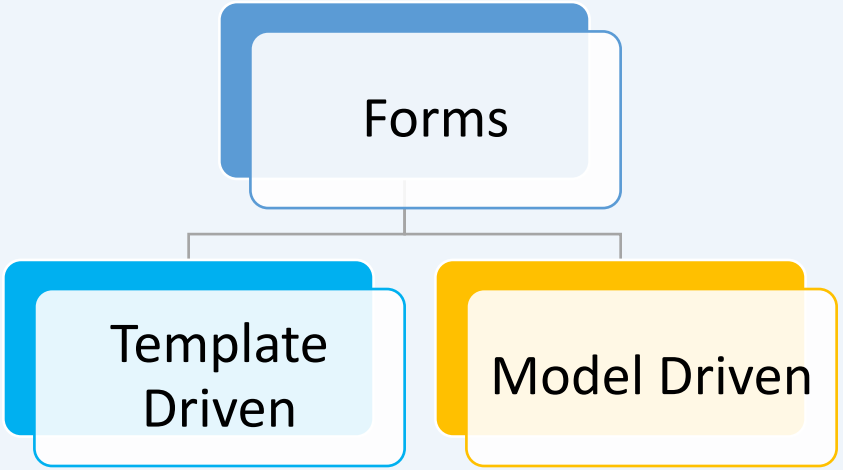
Angular Forms

- HTML forms are an essential part of a web application
- Provides two ways to create form – Template Driven and Model-Driven



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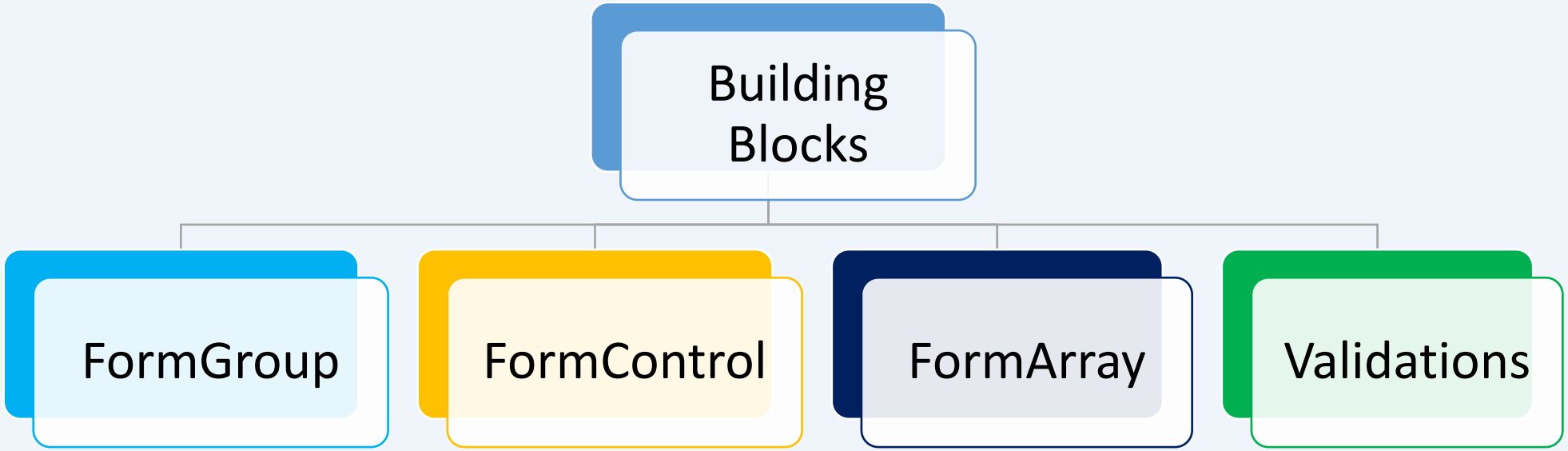
Angular Form Building Blocks



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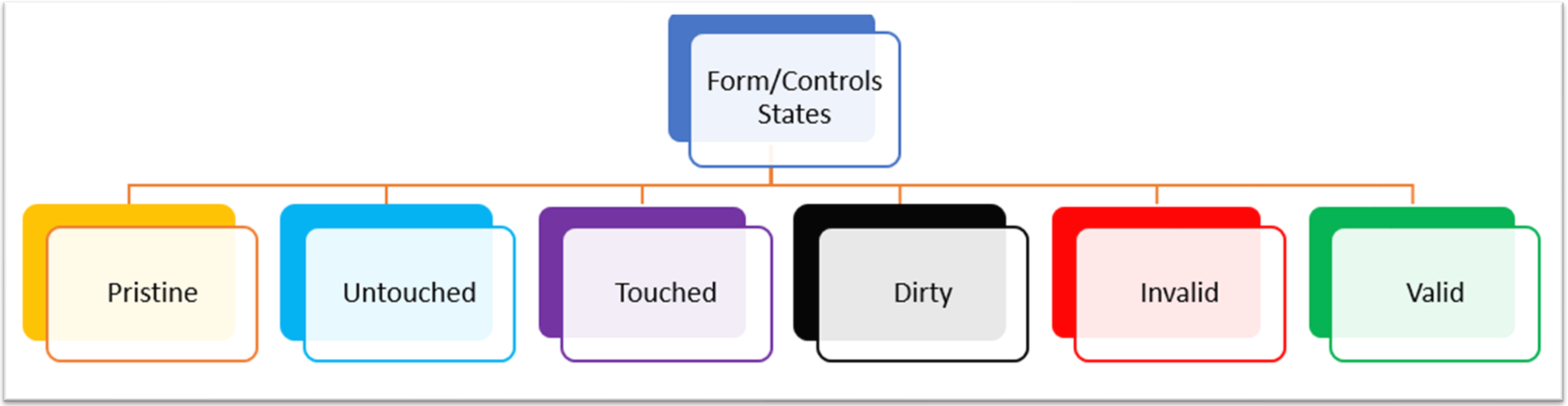
Angular Form and Form Controls States



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States Transition

- untouched => touched
- pristine => dirty
- invalid => valid, valid => invalid

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Template Driven Form

- Form is setup and configured in HTML Code
- Easy and suits for simple form
- Use Directives and Reference Name
  - ngForm
  - ngModel
  - #refName

Form Group

```
<form #myForm="ngForm" (ngSubmit)="save()">
        <input id="name" type="text"
                [ (ngModel) ]="user.Name"
                name="Name"
                #Name="ngModel"/>
</form>
```

Form Control

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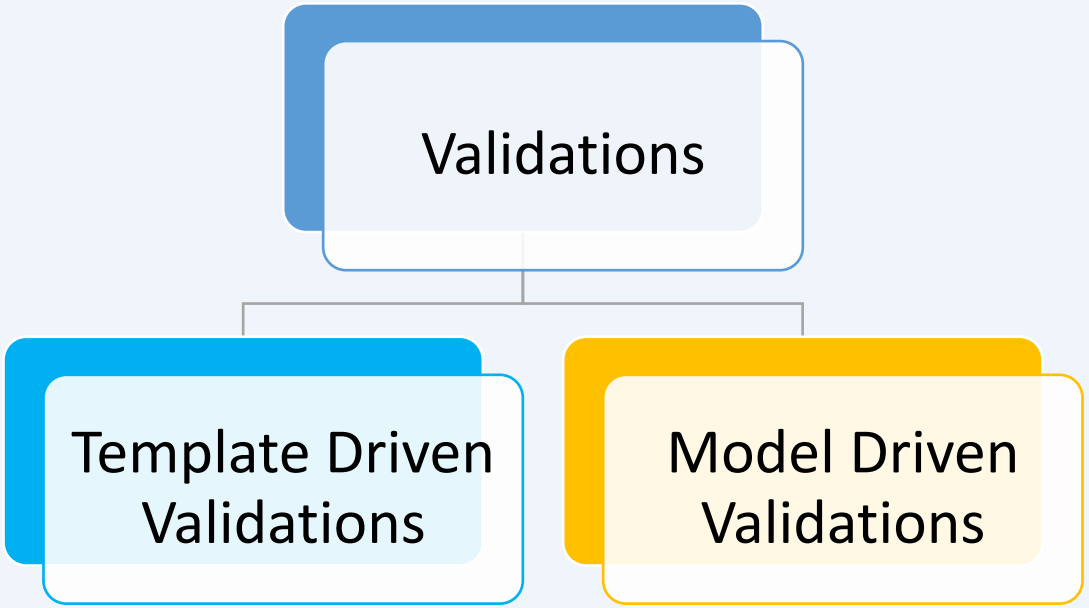
ScholarHat

Angular Form Validations

- Validate user input for accuracy and completeness
- Performs on the client-side.



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Angular Forms Built-In Validation

- required
- minlength
- maxlength
- pattern
- email – supports in Ng4
- min
- max

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Model Driven (Reactive) Form

- Form is setup and configured in the component class
- Based on a *reactive style* of programming where you use the underlying APIs FormControl and FormGroup that track value and validation status
- Reactive forms offer the ease of testing and validation

Form Group

```
<form [formGroup]="myForm">
   <input id="name" type="text"
          formControlName="Name"
          name="Name"
</form>
```

Form Control

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Template Driven vs. Model Driven Form

- Form is setup and configured in HTML Code
- Easy and suits for simple form
- Two-way data binding
- Automatically track form and input element state
- Form is passed to component class via *ngSubmit*()
- Unit Testing is complex

- Form is setup and configured in component class using FormBuilder
- Flexible and suits for complex form
- No data binding(immutable data model)
- Reactive Transformation that can react to changes in data across the app
- Form can be accessed in component class without passing it via ngSubmit()
- Unit Testing is easy

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